

Assessment of Learning Needs and Capacity to Deliver Climate Change Learning in Kenya Report

January 2021



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Acronyms and Abbreviations

ACE	Action for Climate Empowerment
ANOVA	Analysis of Variance
ASALs	Arid and Semi-Arid Lands
AYICC	African Youth Initiative on Climate Change
CCF	County Climate Change Fund
CCU	Climate Change Unit
CCCU	County Climate Change Unit
CCD	Climate Change Directorate
CCCF	County Climate Change Fund
CDM	Clean Development Mechanism
CIDP	County Integrated Development Plan
CoG	Council of Governors
CSO	Civil Society Organisation
DeKUT	Dedan Kimathi University of Technology
DRE	Decentralized Renewable Energy
ESD	Education for Sustainable Development
FAO	Food and Agriculture Organization of the United Nations
FGD	Focus Group Discussion
GCF	Green Climate Fund
ITK	Indigenous Technical Knowledge
JKUAT	Jomo Kenyatta University of Agriculture and Technology
KCCWG	Kenya Climate Change Working Group
KCSAP	Kenya Climate Smart Agriculture Project
KICD	Kenya Institute of Curriculum Development
KII	Key Informant Interview
KPLC	Kenya Power and Lighting Company
KSG	Kenya School of Government
MRV	Measurement, Reporting and Verification
MSC	Master of Science
MTP	Medium Term Plan
MWSI	Ministry of Water, Sanitation and Irrigation
NAP	National Adaptation Plan
NARIGP	National Agricultural and Rural Inclusive Growth Project
NCCAP	National Climate Change Action Plan
NDC	Nationally Determined Contribution
NGEC	National Gender and Equality Commission
NGO	Non-Governmental Organisation
PhD	Doctor of Philosophy
REDD+	Reduce Emissions from Deforestation and forest Degradation
SDG	Sustainable Development Goals
SEforALL	Sustainable Energy for All
SPSS	Statistical Packages for Social Sciences
SRM	Social Risk Management
UNFCCC	United Nations Framework Convention on Climate Change

Chapter

01

Introduction



1.1 Background

Kenya launched the UN CC:Learn Project in 2019 to Strengthen Human Resources and Skills to Advance Kenya's Nationally Determined Contribution (NDC) and the National Adaptation Plan (NAP)

The United Nations Framework Convention on Climate Change (UNFCCC) in Article 6 supports education, training and public awareness on climate change, including the training of scientific, technical and managerial personnel.

Kenya ratified the Paris Agreement in 2016 which highlights the importance of education, training and public awareness to drive climate action.

At national level, Kenya's Climate Change Act (2016) mandates national and county governments to facilitate capacity development for public participation in climate change responses through awareness creation, consultation, representation and access to information and provides mechanisms for and facilitates climate change research and development, innovation, training and capacity building.

Other climate change and education policies including the National Climate Change Action Plan (NCCAP) 2018-2022, the Education for Sustainable Development Policy (2017), also prioritize climate change capacity development as a key enabler to drive Kenya into a low emission, climate resilient development pathway as stated in Vision 2030.

There are also several initiatives and actors within the climate change space that are addressing issues of training and capacity

development within government, civil society, private sector, education and training institutions and media among others.

Since 2011, UN CC:Learn has engaged in Climate Change Learning Programmes with developing countries in Africa, Asia and Latin America, with the purpose of strengthening human resources, climate change learning, and skills development to advance the national climate change development agenda in the respective countries.

During the 2017-2020 implementation phase, the initiative has expanded, following additional support from the Government of Switzerland, which allows support to be provided to new countries, including Kenya.

It is within this framework that Kenya launched the UN CC:Learn Project in 2019 to Strengthen Human Resources and Skills to Advance Kenya's Nationally Determined Contribution (NDC) and the National Adaptation Plan (NAP).

The Project supports the development of a National Climate Change Learning Strategy. The Strategy will be a powerful tool to support the implementation of Kenya's updated Nationally Determined Contribution (NDC) and the National Adaptation Plan (NAP) 2015-2030.

It will strengthen existing processes by taking a more strategic approach to climate change learning.

It will also examine and identify critical learning and skills development needs in key climate-related sectors while addressing challenges in national educational and training institutions and increase opportunities for climate change learning.

The UN CC:Learn's approach will provide added value by contributing to existing processes through an enhanced focus on climate change learning and skills development, supporting the development and implementation of a National Climate Change Learning Strategy aligned with the national NDC and NAP.

The development of a National Climate Change Learning Strategy speaks to a systematic, coordinated approach to climate change learning by fostering multi-stakeholder, cross sectoral collaboration

to identify learning needs and capacity development priorities and strengthen national institutions to deliver climate change learning as part of a broader, sustainable way to enhance an engendered human resource base to address climate change.

1.2 Priority Sectors and Cross Cutting Themes

Kenya has prioritised five sectors and three crosscutting themes in its learning strategy. The sectors and cross cutting themes were agreed on during a national planning workshop that brought together various stakeholders that included the National and County Government, academia and training institutions, private sector, civil society organisations, faith based institutions

and youth and marginalised communities representatives.

The five prioritised sectors are environment, education, energy, agriculture and water, irrigation and sanitation. The three cross cutting themes are public awareness, capacity building and gender and youth engagement.



Figure 1: Prioritised Sectors and Cross Cutting Themes

1.3 Climate Change Learning Assessment Surveys

An important aspect of the National Climate Change Learning Strategy development is to assess learning needs as well as capacities to deliver learning.

The assessment allows an analysis of learning needs as well as creation of a climate change learning action plan for Kenya.

The assessment on climate change learning needs and priority to deliver climate change learning in Kenya started on November 16, 2020 through to December 8, 2020.

Why capacity and needs assessment

1. Identify priorities for learning within and across sectors
2. Inform the identification of actions to be included in the strategy
3. Establish a baseline from which to be able to measure performance
4. Identify existing capacities to deliver learning (institutional and individual)
5. Ensure ownership of key stakeholders in the strategy

Chapter

02

Methodology



2.1 Assessment Methodology

The assessment exercise followed a study approach that promoted the participation of different stakeholders working in the five prioritised sectors namely environment; education; agriculture; energy; and water, irrigation and sanitation.

The assessment also targeted those working in three cross cutting themes namely; public awareness, capacity building and gender and youth engagement. The aim was to enhance the capture of all their inputs into the assessment.

The assessment targeted National and County government institutions with related policy making and implementation functions as well as organisations outside the government with climate change relations operations and functions.

Institutions targeted outside government included; Civil Society Organisation's, Academia and training institutions, media, youth, women and marginalised communities, private sector and development partners

The assessment also relied on Annex 2 and 3 of the UN CC:Learn guidance note on developing a National Climate Change Learning Strategy which has questionnaires for assessing climate change learning needs and priorities and capacity to deliver climate change learning.

The assessment kicked off on November 14, 2020 and concluded on December 8, 2020. Both qualitative and quantitative methodologies were employed to collect data using participatory approaches.

2.2 Assessment Phases

The assessment followed three phases as indicated in Figure 2.

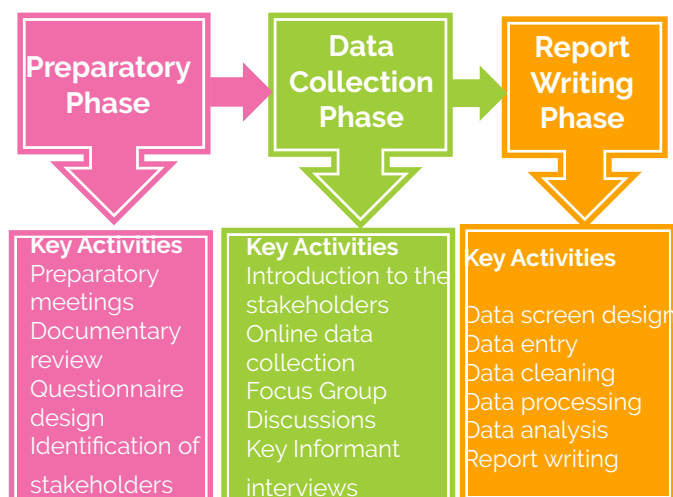


Figure 2: Assessment methodology phases

2.3 Assessment Approach

The following three approaches were used in the assessment;

- 1 Online questionnaire
- 2 Focus Group Discussion (FGD)
- 3 Key Informant Interviews (KII)

Online questionnaire

To facilitate stakeholder engagement and streamline data collection, an online data collection tool (Kwik Survey) was used to facilitate data consolidation and reporting by key stakeholders. The online platform was used as it is efficient, interactive and user friendly. Two online survey were developed namely:

1. **Survey 1:** Assessing individual climate change learning needs and priorities in Kenya <https://kwiksurveys.com/s/vj5eDXGF>
2. **Survey 2:** Assessing institutional capacities to deliver climate change learning in Kenya <https://kwiksurveys.com/s/gsYsILhs>

A total of 210 individuals with related policy making and implementations functions as well as climate change-related operations and activities undertook survey 1 while 59 institutions that provide some form of climate change training undertook survey 2.

Focus Group Discussion (FGD)

A total of 9 FGDs were held that brought together 238 stakeholders drawn from the National and County Government, Civil Society Organisation's, Academia and training institutions, youth, women and marginalised communities, private sector and development partners.

Data Consolidation and Analysis

This process involved inspecting, cleaning, transforming, and modelling data with the goal of extracting relevant information, informing conclusions and supporting decision-making.

This took into consideration multiple facets and approaches, encompassing diverse techniques to ensure all the key issues raised by the respondents in different questionnaires were captured.

Frequency counts of responses were obtained so as to generate descriptive information about the respondents who participated in the assessment and to illustrate the general trend of findings on various variables that were under investigation.

Qualitative data was analysed thematically along with the study objectives and presented thematically in narratives forms whereas the quantitative data were analysed descriptively using frequencies and percentages and inferentially using analysis of variance (ANOVA) against binary regression in Statistical Packages for Social Sciences (SPSS).



Chapter

03

Assessment of Learning Needs and Priorities



3.1 Survey 1 Target Audience

Survey 1 on assessing individual climate change learning needs and priorities in Kenya targeted individuals with related policy making and implementations functions as well as climate change-related operations and activities.

The survey was meant to obtain an understanding of their learning and skills development priorities.

3.2 Respondents Gender

Men made up the majority of the respondents at 66% while women were 34%

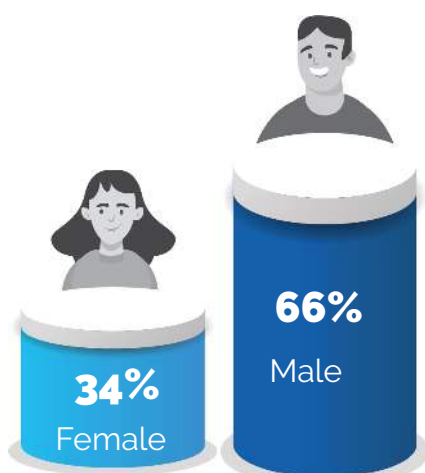


Figure 3: Percentage of male and female survey 1 respondents

3.3 Persons With Disability

Only 4% of the respondents indicated that they are Persons with Disability. The main disability is partial hearing and physical impairment.

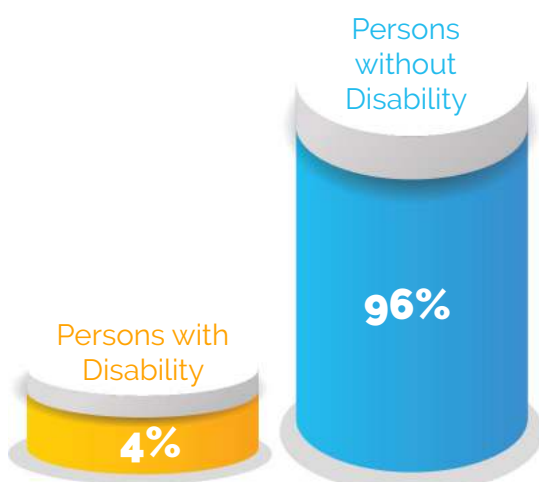


Figure 4: Percentage of persons with disability in survey 1

3.4 Respondents Age Group

Youth consisted of 48 % of the respondents while 2% were aged 60 years and above. A further 27% were aged 36-40 years while 23% of the respondents were between 41-60 years.

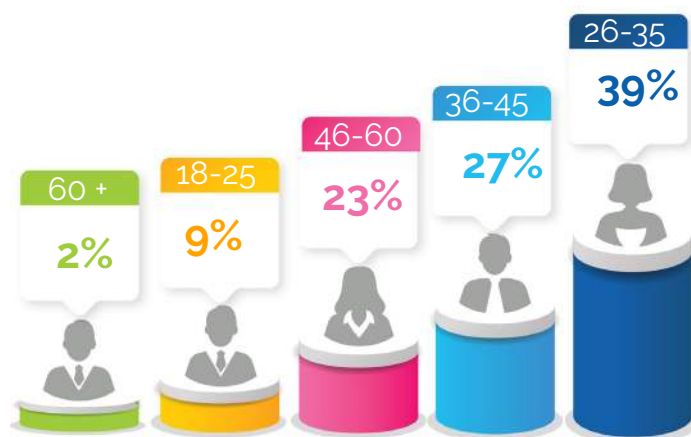


Figure 5: Survey 1 respondents age group

3.5 Type of Institution

County Governments had the highest respondents at 29% followed by Civil Society Organisations and the National Government at 28% and 18% respectively. The media had the lowest respondents at 2%.

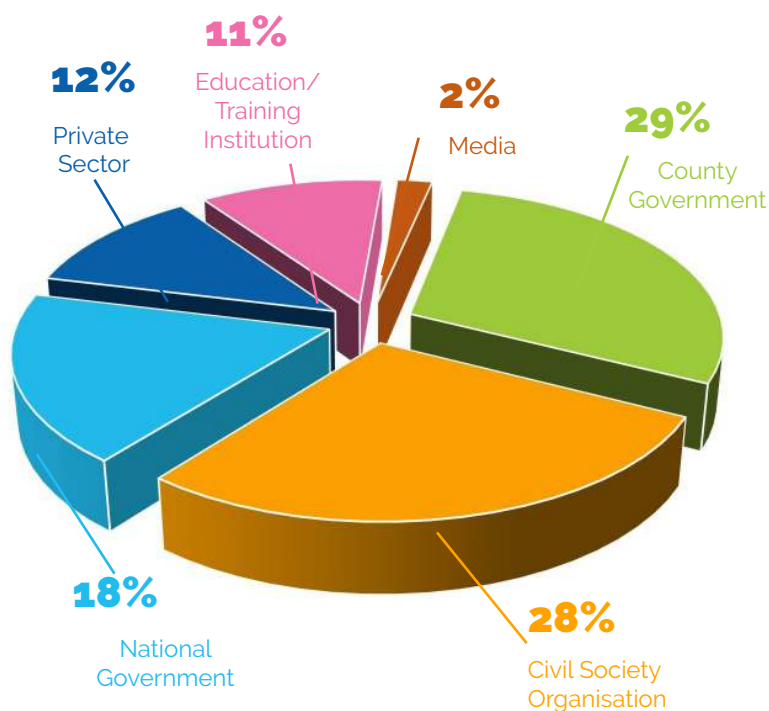


Figure 6: Institution representation

3.6 Sector and Cross Cutting Theme

The environment sector had the highest number of respondents at 137 followed by the cross cutting theme capacity building at 67, public awareness at 61, water, irrigation and sanitation at 56, agriculture at 49, energy at 47, gender and youth engagement at 45 and education at 38.

Health and public finance management each had 2 respondents while building and construction sector had one respondent.

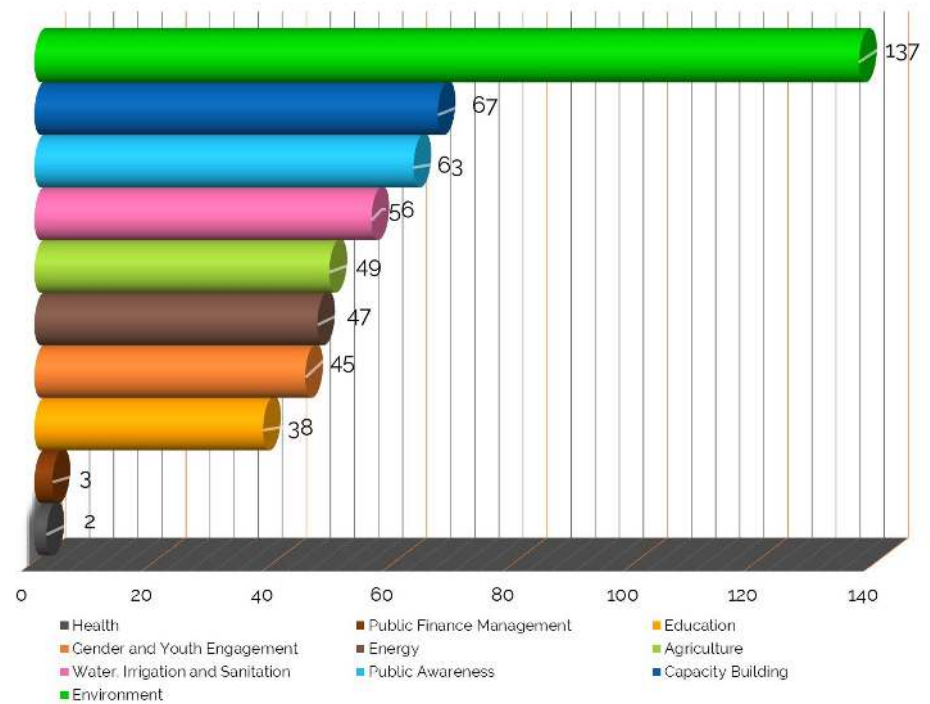
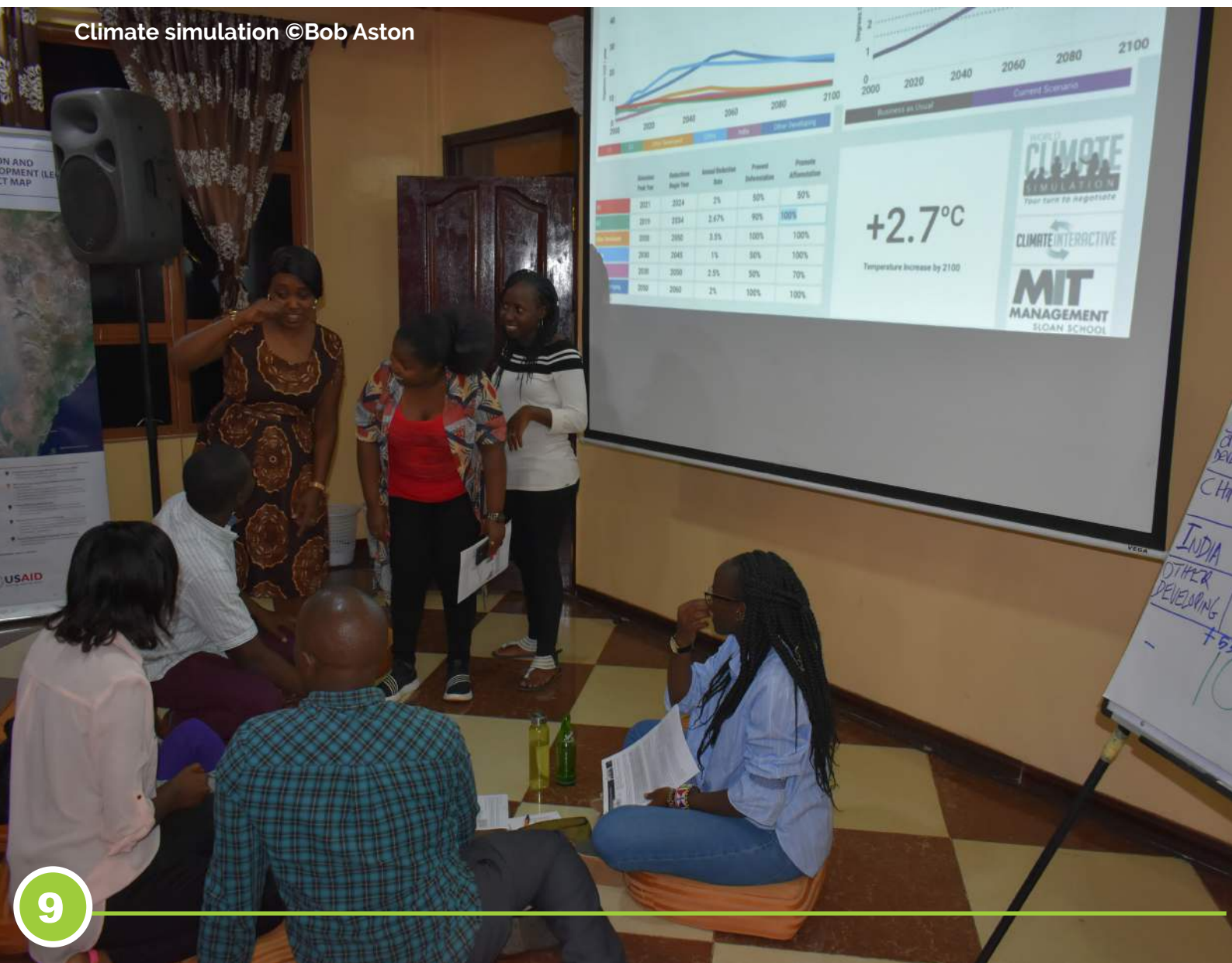


Figure 7: Sector and crosscutting theme representation



3.7 Climate Related Functions and Tasks

The respondents listed the following as the specific climate change related functions and tasks that they perform in their institutions.

1. Training farmers and stakeholders on climate change impacts and technologies/techniques to adopt to mitigate e.g conservation agriculture
2. Development and improvement of climate smart agriculture technologies
3. Knowledge generation and dissemination
4. Capacity building various stakeholders on climate change adaptation and mitigation
5. Researching on specificity of climate change adaptation across various livestock production systems of Kenya, Ethiopia and Uganda
6. Developing protocols for long-term monitoring of adaptation in livestock sector
7. Training policy makers and building their capacity to monitor, report on adaptation to the UNFCCC and use the information in adaptation planning
8. Climate amelioration through tree planting
9. Research that informs evidence generation and use
10. Regenerative agroforestry, permaculture/ agroecology model of farming
11. Appropriate water conservation techniques
12. Sustainable control of invasive plants and sustainable utilization of non wood based products
13. Advocacy on renewable energy, clean cooking technologies, communicating climate change, documenting and disseminating climate change adaptation actions
14. Water harvesting through sand dam and shallow well construction
15. Reforestation, agroforestry, rain water harvesting skills enhancement
16. Capacity building on quality seed and appropriate seeds to use for planting as per ecological areas
17. Rehabilitation of abandoned traditional foods by farmers
18. Agroecology to enhance soil, animal and human health for agricultural and environmental well being
19. Evaluating campus energy and waste management practices as well as giving recommendations to the relevant authorities
20. Creating community awareness within the campus and neighboring areas on campus energy and waste management
21. Forest development services such as, tree nursery establishment, school greening Program, promotion of on farm tree planting and promotion of private tree nurseries
22. Capacity building of county governments and communities on mainstreaming climate information in planning and implementation
23. Support to the Natural Resource Management Committee on matters mainstreaming climate change in the education sector
24. support the development of greenhouse gases (GHG) inventory
25. Support the development of guidelines for mainstreaming climate change into all learning curriculum
26. Support the development of an MRV+ system
27. Implementation of CDM projects, Partnerships for implementation of Paris agreement, Sale of CERs from CDM projects and carbon footprint analysis

3.8 Required Climate Change Knowledge and Skills

When asked about the knowledge and skills required to perform specific climate change related functions and tasks at their institutions, the respondents listed the following:

1. Understanding of climate change basics
2. International efforts put in place to mitigate the effects of climate change
3. Indepth understanding of global climate change framework
4. Stakeholder mapping and engagement for MRV+ Reporting
5. Skills on calculation of emission factors
6. Data management, GIS and remote sensing skills
7. Knowledge in developing indicators for mitigation and adaptation reporting
8. Software applications for climate change mitigation and adaptation
9. Data collection and integration
10. Climate change negotiation skills
11. Monitoring and Evaluation of country commitments on climate change
12. Climate change monitoring, evaluation and reporting skills
13. Climate change financing and budgeting
14. Designing projects that are climate change compliant and those that can attract funding
15. Knowledge on climate change adaptation
16. Negotiation skills on climate change policies and programmes
17. Fund raising and resource mobilisation skills
18. Understanding of natural water cycle
19. Knowledge in the appropriate mode of delivery of climate change courses
20. Knowledge on importance of soil in biodiversity support,
21. Climate change mitigation strategies in value chain financing
22. Knowledge and skills in the best pedagogy for teaching aspects of climate change
23. Interrelatedness of environmental elements (Flora, fauna, Human)
24. Knowledge of the interaction but climatic and non-climatic pressures at the local level,
25. Knowledge of policy and institutional context within the country
26. Climate change training facilitation skill
27. Climate change assessment tools
28. climate modeling and forecasting
29. Carbon footprint analysis
30. Qualitative climate data analysis
31. Methods of carbon sequestrations and trading
32. Understanding of greenhouse gases and policies that can be used to set up E-mobility infrastructure in Africa
33. Agroforestry, regeneration of local tree species and nursery skills
34. Assessment of climate change risks
35. Importance of soil in biodiversity support, Interrelatedness of environmental elements (Flora, fauna, Human)
36. Understanding of natural water cycle and relationship between invasive plants and rangeland health.
37. Knowledge on renewable energy and clean cooking technologies
38. Policy and advocacy against climate change at county and national government
39. Policy design and development
40. Periodical weather forecast alterations, crop cover for enhancing soil conservation as well as ecological zone favorable crops for food as well as nutritional attainment and livelihood improvement.
41. Technology, innovations and management practices initiatives
42. Climate Change Mitigation and Management skills related to Kyoto Protocol
43. Knowledge on Article 6 of the Paris Agreement on market mechanisms
44. Knowledge on bioenergy development and promotion of clean cookstoves
45. Understanding of the relationship between climate change and health
46. Knowledge of how the total abatement of GHG in the energy sector is arrived at and the potential activities that are considered
47. Knowledge on energy sources and impacts on climate
48. Knowledge on renewable energy and clean cooking technologies
49. Knowledge on waste management practices and impact on climate

50. Knowledge on climatic processes

51. Ecosystem Services Analysis

52. GHG inventory development skills

53. Carbon accounting skills

54. Skills to perform socio-economic risk assessments of climate change impacts that can be used by to inform its actions and plans

55. Knowledge on effects of temperature and precipitation changes on both biotic and abiotic environment

56. Understanding of climate smart technologies and technological needs of developing countries

57. Relationship between invasive plants and rangeland health

58. Integrated climate risk management and disaster risk management

59. Solutions to food waste and inventory activities to enhance reduction of food waste

60. Research, innovation and entrepreneurship in climate change

61. Knowledge on climate change governance

62. Skills in dry land forestry management and restoration of degraded lands

63. Knowledge on the role of Innovation in reducing emissions

64. Solutions to food waste and inventory activities to enhance reduction of food waste

65. Anticipatory decision making for climate action

66. Global land use information and mitigation on land and the SDGs

67. Knowledge on how to intergrate sustainable methods in the education sector

68. Design and fabrication skills

3.9 Training Skills Development Activities

When asked whether they have received any training or skills development activities in the past 2 years, 50% of the respondents indicated that they have received while a further 50% indicated that they have not received any training.



Figure 8: Training skills development activities

Some of the skills development training received include

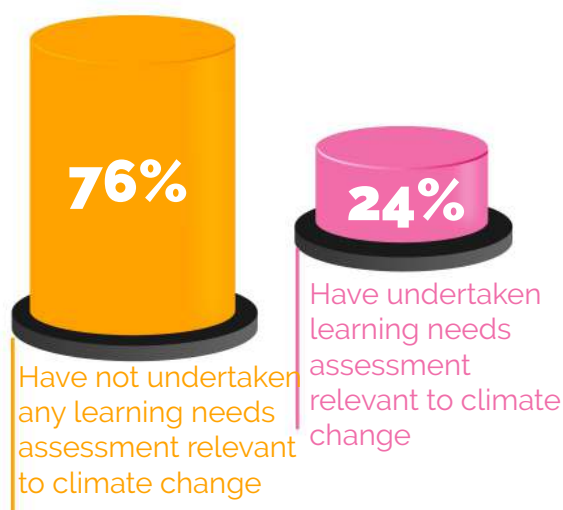
1. Training on climate smart agriculture
2. Climate governance, diplomacy, negotiations and leadership training
3. Training on decentralised renewable energy and clean cooking technologies
4. Cost-Benefit Analysis of climate actions training
5. Undertaking MSc course in climate change and sustainability
6. Climate change expert training on the role of solar technologies in climate change adaptation in small scale agriculture
7. Training on atmospheric processes and climate change
8. Training on designing programs that are climate change compliant
9. Training on access to climate finance and mainstreaming of climate change in planning and budgetary process
10. Training on mainstreaming climate change actions on CIDPs
11. Training on climate change policy, planning and budgeting for Climate Change Units (CCU)
12. Skills training on Implementation of CDM Projects at KenGen
13. Training on how to inform the public on climate change as a community media organization
14. Training on Building resilience by mainstreaming climate change actions into the businesses

15. Journal writing on guidelines of Climate Change News Reporters
16. Certificate in Education for Sustainable Development (ESD)
17. Certificate in Making Research Work: Solving complex agri-food problems through action research
18. Certificate in Fundamentals of Information Literacy and Access to Global Online Research in Agriculture (AGORA)
19. Climate change reporters and bloggers training
20. Global financial mechanisms for Climate Governance training
21. PhD in Climate Change and Adaptation
22. Undertaking MSc in climate change and adaptation
23. Climate change mitigation measures training
24. Training on communicating climate change in digital platforms
25. Training on development of National Determined Contribution (NDCs) Monitoring tools
26. REDD+ readiness academy training
27. Social Risk Management (SRM) course
28. Environmental Impact Assessment (EIA) course
29. CSOs engagement mechanism on climate adaptation and mitigation
30. Training on gender mainstreaming in climate change actions
31. Empowerment of IPs on climate governance and policy development
32. Capacity Building on Action for Climate Empowerment (ACE)
33. Online course for school teachers to be accredited by UNFCCC
34. Specialized Module on cities and climate change
35. Training on air quality monitoring and reporting
36. Climate city planner course and cross sectoral climate change linkages with action prioritization tools
37. Training on climate actions and access to the Global Green Climate Fund (GCF)
38. Training on climate resilient water management
39. Training on climate change laws and policy formulation skills
40. Training on resource negotiation and contextualisation of climate change impact
41. Climate change proposal writing, budgeting and monitoring
42. Online course on climate change adaptation and mitigation
43. Training on updating water sector NDC on water and climate change
44. Undertook an course on Climate Action: Solutions for a Changing Planet
45. Training on County Climate Change Fund (CCCCF) Mechanism
46. Vulnerability assessment training



3.10 Learning Needs Assessment

When asked whether they have undertaken any learning needs assessment relevant to climate change, 24% of the respondents said that they have while 76% have not undertaken any learning needs assessment relevant to climate change.



Among the assessment undertaken include;

1. Training needs assessment for county government officials that were involving in designing and implementing the County Climate Change Fund (CCCF) mechanism
2. Capacity Assessment for the CCU of the Ministry of Water, Irrigation and Sanitation which was used to develop a Capacity Development Strategy and Action Plan for the CCU
3. County climate change mapping and county climate change risk profile assessment
4. Training needs assessment for members of Clean Cooking Association of Kenya; Kenya National Commission for UNESCO (KNATCOM) Youth; and the built environment professionals
5. Evaluation of impact of sector development plans
6. Training on climate change financing and mainstreaming of climate change into planning and budgetary process
7. Capacity needs assessment for CSOs and county governments

8. Kenya Climate Smart Agriculture Project.
9. Women and the role of indigenous knowledge to climate change action
10. Climate change policy and bill development
11. Participated in a survey of an expert carrying out county climate profile
12. Paris Agreement conversance in Kenya
13. Assessment of institutions and the organization on climate change skills
14. Carbon exchange/credit
15. Flood early warning
16. Carbon finance, carbon footprint analysis
17. Skills assessment by IEWM Gender and Climate Change Working Group Kenya
18. We deliberate internally and share with our development partners staffs' learning needs on an annual basis which then informs our annual capacity building sessions
19. Global warming reduction
20. On capacity enhancement and stakeholder engagement
21. Need for knowledge on Climate Finance, negotiation skills, development of bankable climate/ green development proposals

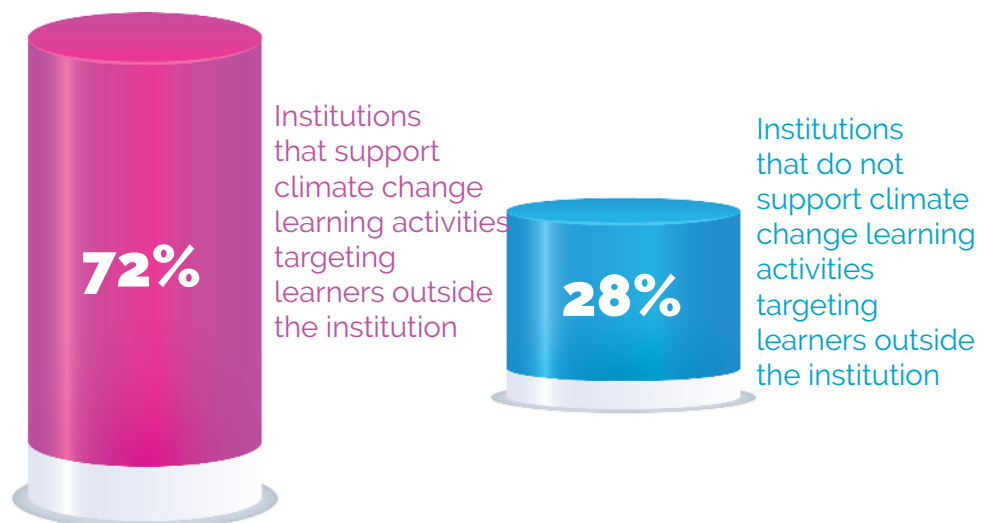




3.11 Targeting Learners Outside Institutions

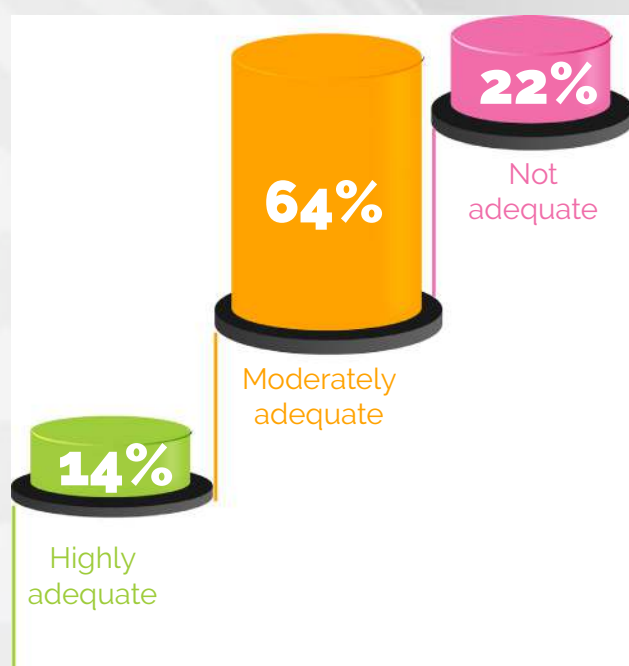
Seventy two percent of the respondents institutions offers projects or programmes that support climate change learning activities targeting learners outside their institutions.

Similarly, 28% of the institutions do not offer any programme or project that targets learners outside the institution.



3.12 Human Resource Capacities

When asked to what extent are human resource capacities and skills levels in their institution sufficient to enable them perform their climate change-related functions and tasks, only 14% of the respondents said that its highly adequate. A further 64% said that it is moderately adequate while 22% said that it is not adequate.



Some of the reasons shared include

Education

- Climate change learning is not embedded in curriculum of Kenya's education system
- The Ministry of Education does not have a Climate Change Unit
- Ministry of Education staff have not been trained on climate change
- Climate change is not taught by many institutions of higher learning
- Limited knowledge in mainstreaming climate change concepts in curriculum designs and designing learning experiences on climate change concepts
- Many institutions of higher learning are not implementing green or clean energy systems in their daily campus activities
- Inadequate research support to inform what should be in the curriculum at the university level
- Insufficient staff, resources and inadequate appropriate strategies for addressing climate change
- Financial constraints in the roll-out of curricula or development of more curricula and conducting research on climate change
- Low level of climate change training through all levels of education from primary to tertiary education
- Climate change is currently not a priority recruitment and training area since it does not contribute to the key mandate of the Ministry of Education
- Weak link/collaboration between the Commission for University Education and researchers on research outputs to inform curriculum development, review and reforms
- Inadequate training of technical staff of the Commission for University Education and peer reviewers (mostly academia and industry) on recent trends in climate change matters and their relevance so that as they evaluate curriculum
- In approving/inspecting universities for accreditation/quality assurance, it is expected that the Commission for University Education ensures that energy saving, use of renewable energy, environmentally friendly approaches in constructions, green economy among others are mainstreamed. This is currently not happening and the Commission needs to be capacity built to support this function

Environment

- Lack of an in-house meteorologist and environmental economist with solid experience in the development of climate scenarios (for the next 10, 20, 25, 50 Years) and climate finance modelling at the Ada Consortium
- Inadequate knowledge on climate change impacts
- Climate change is deemed to be

synonymous with environmental issues which may not be the case

4. There is limited and inadequate co-ordination of climate change activities between the relevant Ministries
5. Inadequate partnerships with other Ministries and other stakeholders to allow for effective coordination in implementing the NAP and NCCAP
6. Challenge in supplying of digital gadgets to monitor forest use

Energy

1. Most of the staff in the bioenergy division of the Renewable Energy Directorate have not been actively engaged in monitoring and reporting on climate change related data much as they are able to identify actions that contribute to the reduction of GHG emissions
2. Insufficient technical knowledge from stakeholders on E-mobility concept
3. Limited knowledge on affordable renewable energy and clean cooking technologies
4. Limited knowledge and skills on green technology jobs and economy
5. Inadequate skills on wind energy harvesting infrastructure and large-scale solar power generation system and sale agreement with KPLC
6. Inadequate training to update skills, due to inadequate funds

Water, Irrigation and Sanitation

1. Inadequate human resources. On the establishment of the CCU, staff were not appointed to work specifically in the Unit. This leads to climate change actions to not be prioritized, unless climate change is included in performance contracts
2. Although CCU Desk Officers have officially been appointed at the Ministry of Water, Irrigation and Sanitation, there is no financing yet to carry out the mandate. The deployed officers do not have the equipment and have not been trained to carry out their duties

3. The CCU at the Ministry of Water, Irrigation and Sanitation requires funds to effectively implement the priority actions listed in the NAP. However, it does not receive funds that specifically support climate change projects
4. Currently, different types of water (e.g., blue water, black water and grey water) are handled by different Ministries hence effective coordination is a challenge
5. Inadequate institutional framework/structure to support climate change in the Ministry of Water, Irrigation and Sanitation
6. There has been inadequate investment to support knowledge management in the water sector
7. Inadequate infrastructure for water treatment, waste-water recycling, rain water harvesting and water storage

Agriculture

1. Low understanding of the frameworks especially related to agriculture
2. Lack of financial and human resources to address climate change
3. Those in agriculture do not have a clear understanding of climate change adaptation and mitigation strategies
4. Limited knowledge on hydroponics and aquaponic production by use of green energy powered water reticulation systems
5. Privately generated climate smart initiatives which often are documented but unavailable

Capacity Building

1. Capacity building for climate change has not been identified as a training need at the institutional level
2. Low knowledge on monitoring and evaluation skills especially for mitigation investments
3. Inadequate training capacity and experience due to lack/inadequate funding/scholarship and limited exposure
4. Overreliance on external consultants to undertake climate change training

5. Current staff have no formal training on climate change
6. Limited capacity on climate change policy design and legislative drafting
7. Lack of formalized training and interactive, socially diverse material accessible at a community level
8. Implementing policies on climate change mitigation at the county and national level has been a challenge
9. Limited training on climate change reporting
10. Inadequate capacity building on development of market mechanisms procedures and methodologies required for successful implementation of climate change mitigation projects
11. Inadequate training on climate finance and climate trends
12. Inadequate resources to build on existing capacity
13. Staff not fully equipped on matters of climate change
14. Gaps in organizing training due to inadequate human and financial resources
7. Limited knowledge on climate monitoring and indicators of climate change
8. Lack of knowledge brokers to translate climate change science knowledge into local solutions
9. Limited knowledge on climate change mainstreaming at organizational and community level
10. Limited understanding of climate financing and GCF readiness
11. Low knowledge on climate change causes and impacts and identification of effective adaptation and mitigation measures to climate change effects

Gender and Youth Engagement

1. Limited knowledge on gender responsive budgeting
2. Difficulty in communicating climate change to a dynamic youth audience
3. Few youth specific resources on climate change
4. Climate change capacity needs to start from the young generation so it can be part of their culture as they grow in turn creating a climate smart generation
5. Limited transformative collective action to ensure women play a key role in climate change issues at the local level
6. How to mainstream gender in climate change budgeting and reporting

Public Awareness

1. Poor response to climatic change due to higher level of illiteracy among the community and ignorance
2. Insufficient knowledge around climate change adaptation and mitigation due to lack of needed experience, funds and mentorship
3. Unavailability of simplified information packaged for stakeholders to interpret the likely impacts of climate change on their livelihoods
4. Lack of knowledge and skills on climate change amongst majority of the staff
5. Poor information sharing from lead Government agencies on climate change achievements and failures
6. Limited knowledge on community engagement on utilising scientific climate information and along with indigenous information for production

General Issues

1. Lack of urgency in addressing climate change matters thus leading to insufficient allocation of resources to equip officers with knowledge and skills necessary to combat climate change
2. Climate change is not accorded the due weight in many institutions of government
3. Insufficient staff, resources and inadequate appropriate strategies for addressing climate change
4. Limited knowledge and skills on qualitative climate data analysis, methods of analysing carbon sequestration and

modelling

5. Inadequate climate change negotiations skills
6. Inadequate finances to sustain advocacy actions
7. Inadequate training and exposure to institutions dealing with climate change related issues
8. Overlap of functions and responsibilities which can cause the activities carried out by one Unit to be undermined despite their relative importance
9. Lack of skills on climate change financing and development of bankable projects
10. Being 100% donor funded, staff costs are pegged on projects hence trained staff turn over
11. Donor driven policies i.e., when fund is not available it stagnates the implementation process

12. Difficulty in working with some of the County Government's makes it difficult to undertake some climate actions
13. Limited mechanisms for strengthening consultation and participation at the policy level
14. Lack of expertise in climate project pipeline development
15. Unavailability of tools for capturing climate change data
16. Limited understanding of the approaches and strategies for transparent reporting of climate change mitigation and adaptation and communication pathways for reporting
17. Gaps in policy analysis, communication, political leadership, ecosystem approach, climate and policy skills
18. Limited knowledge and skills in mainstreaming of climate change in development of policies, economic modelling and labour issues

Measurement of GHG emissions from the agricultural sector ©LECRD Project



3.13 Climate Change Topics Learning Priority

When it comes to cross cutting climate change topics that are relevant to respondents in their role within the various institutions

Learning how to communicate about climate change and learning about green economy growth and jobs had the highest priority at 73% each.

Learning about adaptation and climate resilience; learning about mitigation and emission reduction strategies; learning about climate change and disaster risk management and learning about the role of youth in climate change action each tied at 70%.

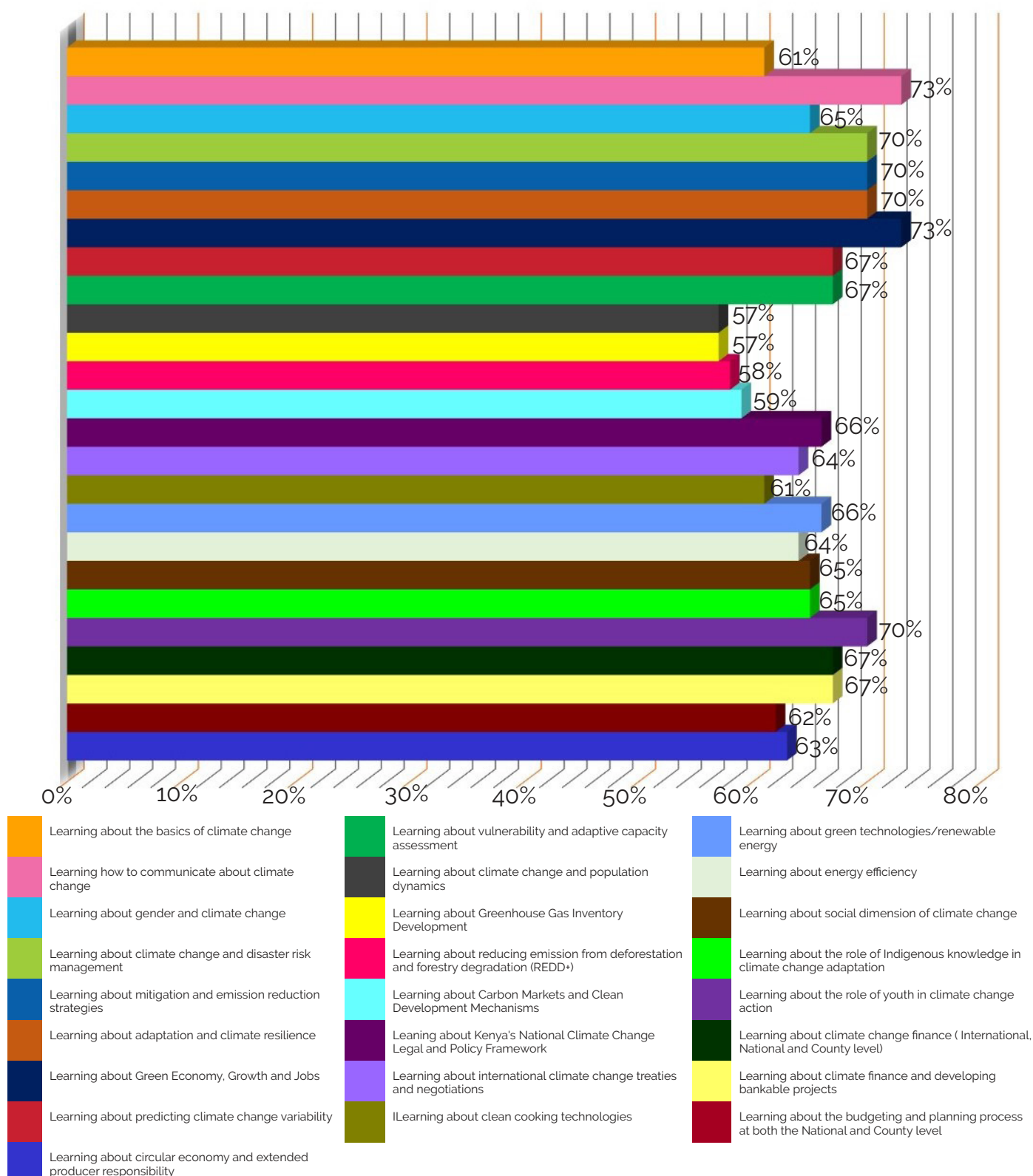


Figure 9: Climate change topics learning priorities

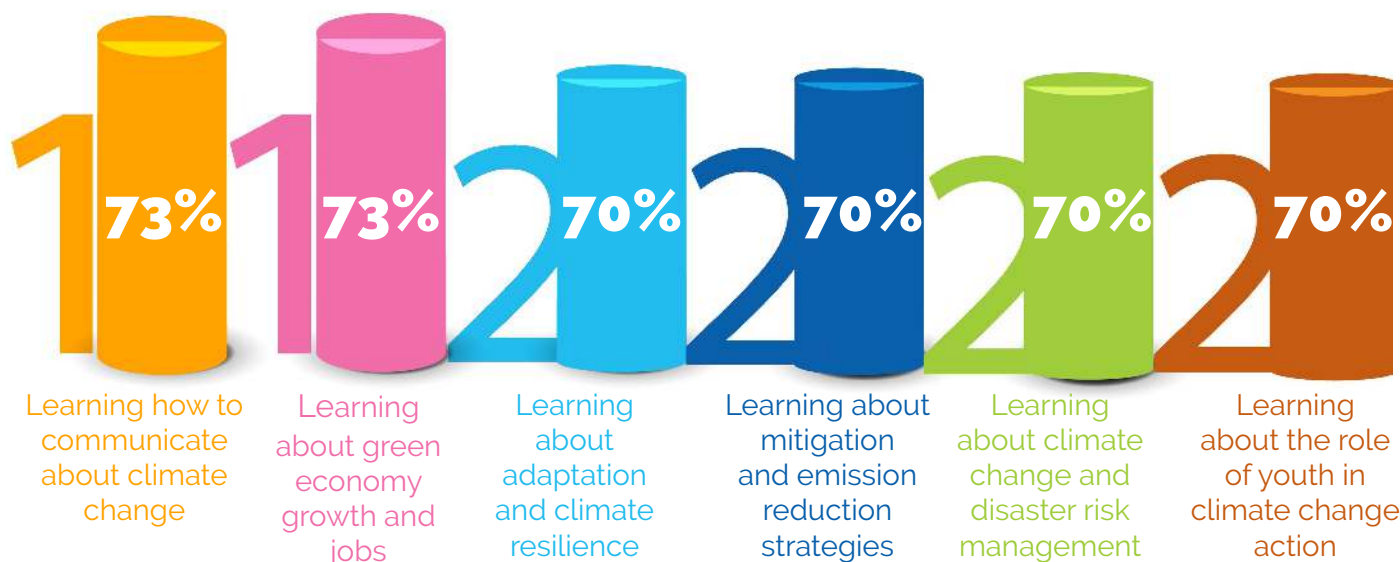


Figure 10: Prioritised learning topics by respondents

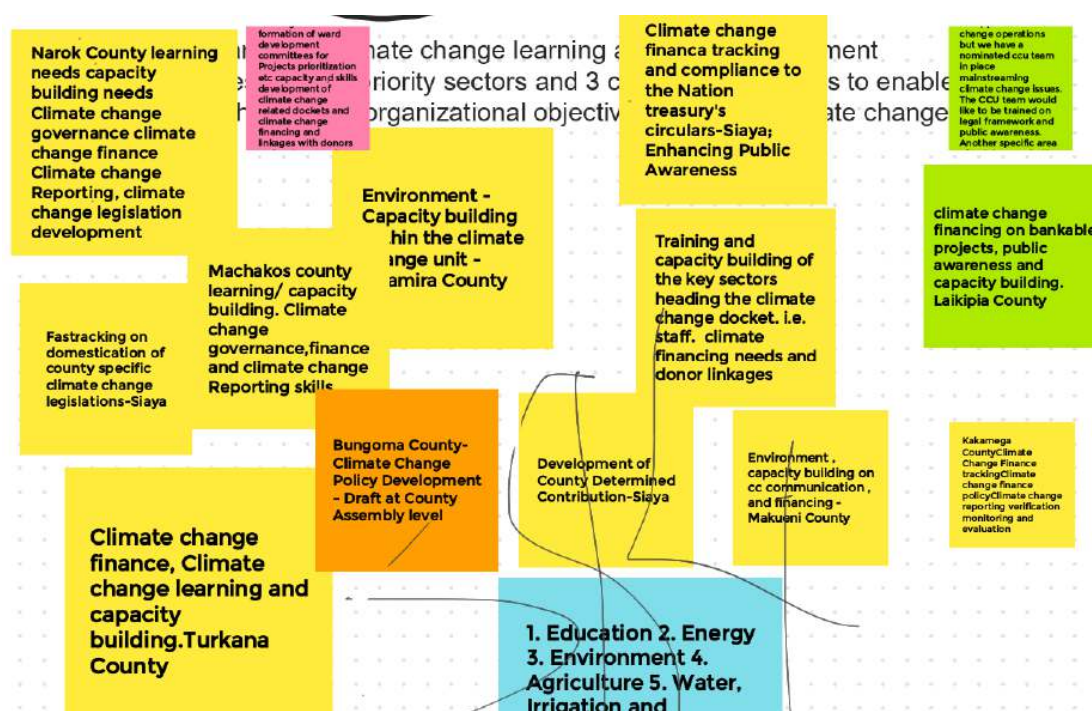


Figure 11: Some of the County Governments Learning priority areas



Figure 12: Some areas where County Governments require capacity building support

Chapter

04

Assessing Institutional Capacity to Deliver Climate Change Learning



4.1 Survey 2 Target Audience

Survey 2 on Assessing Institutional Capacities to Deliver Climate Change Learning in Kenya targeted institutions that are already engaged in, or might play a potential role in, delivering climate change learning actions in the five prioritized sectors and three cross cutting themes.

The survey was meant to elicit information on existing institutional capacities to deliver climate change learning actions and to explore opportunities for scaling up delivery of climate change learning through national institutions.

4.2 Institutional Classification

When asked how their institution can be classified, 34% said that they are from Civil Society Organisation, 18% Professional Learning and Skills Development, 13% Specialized training institution, 11% Basic Education, 9% Textiary Education, 7% Higher Education and Training, 5% Research institute, 2% Technical and Vocational Education and Training and 2% Media.

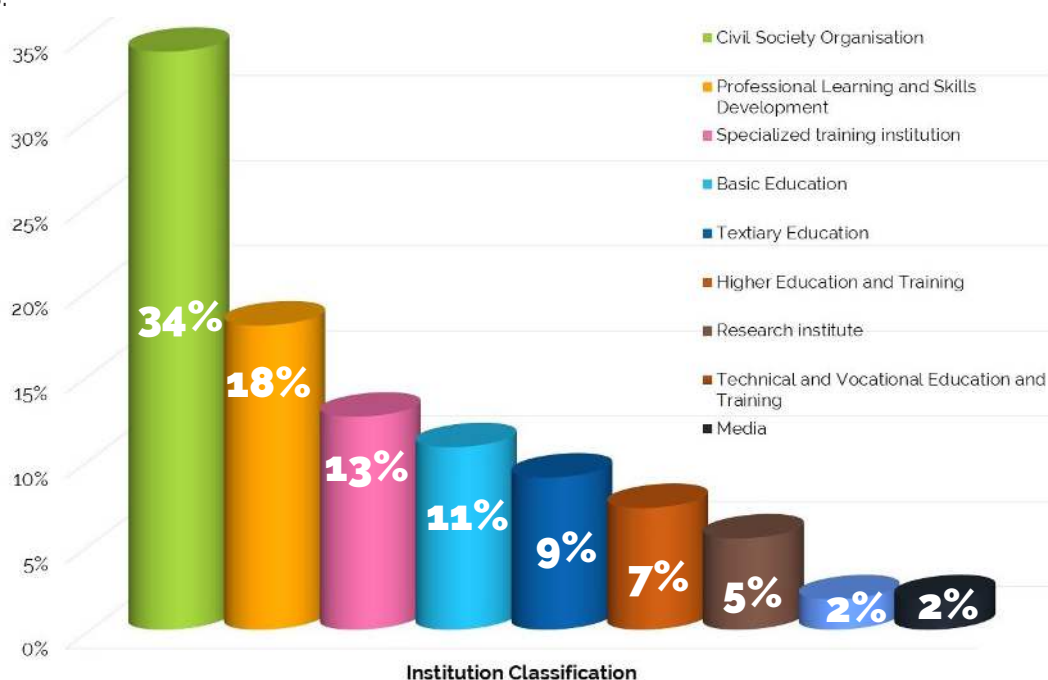
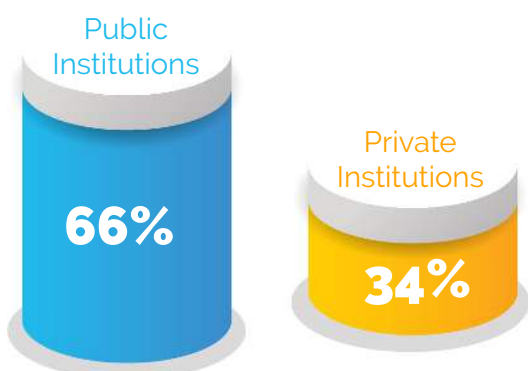


Figure 13: Survey 2 respondents institutions

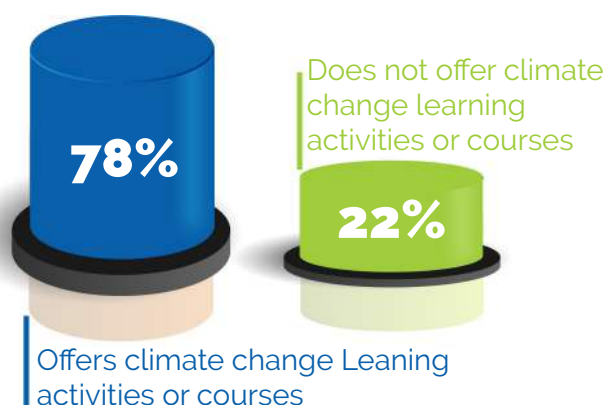
4.3 Institution Registration Status

When asked whether their institutions are public or private, 66% of the respondents said that their institutions are public while 34% said that they are in private institution.



4.4 Climate Change Learning Activities

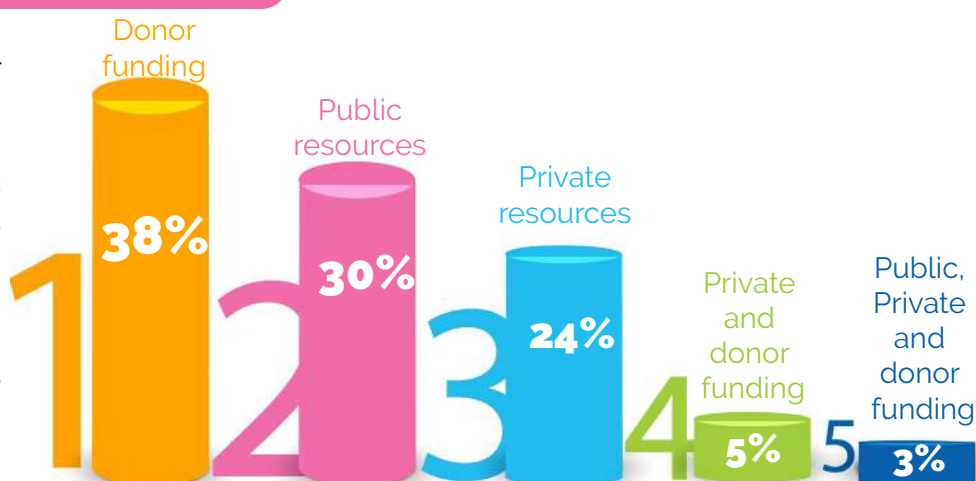
When asked whether their institutions offer learning activities or courses on, or related to, climate change, 78% said yes while 22% said that they do not offer.



4.5 Learning Activities Financing

When asked how learning activities delivered by their institutions are financed, 38% said that they rely on donors followed by public and private resources at 30% and 24% respectively.

An additional 5% rely on both private and donor funds while 3% rely on public, private and donor funding.



4.6 Programs Appropriate for Persons With Disability

When respondents were asked whether they offer any climate change related training programs that are appropriate for persons with disabilities, 44% indicated that they offer while 56% do not offer any climate change training that is appropriate for persons with disability.



4.7 Climate Change Related Training Participants

According to the respondents, an average of 21,685 people have annually been participating in climate change related learning activities,

Although most of the respondents did not share gender disaggregated data, those who shared indicated that on average 55% of the beneficiaries are male while women make up 43% of the beneficiaries with persons with disability consisting of only 2% of the beneficiaries.

Some of the training activities include

- Exposure and learning visit to successful climate change projects
- Visits to Mazingira Centre and low emissions livestock practices
- Tailor made courses on establishment of the County Climate Change Fund (CCCF) mechanism
- Creating a joint platform of activists for Action in Kenya
- Offering a platform for research(social, ecological, community) as to inform decisions at different levels
- Collaborating with other organizations(Local/International) in implementing local action
- Capacity building on traditional knowledge relevant for climate change resilience ,gender and climate change, basics on climate change among others
- Fostering and enhancing environmental education in schools and communities across Kenya
- Engaging in sustainable food practices and ecological farming principles
- Climate smart practices in combating the climate change pact for sustainable productivity and profitable undertaking in agriculture
- Community awareness raising on climate change and food security, causes and effects of climate change on the

- environment and mitigation measures
12. Training communities on climate governance and green energy
 13. Agroecology for both adaptation and mitigation
 14. Training stakeholders on climate change adaptation techniques and mitigation technologies
 15. Capacity building stakeholders on the role of indigenous People in climate change discourses
 16. Training workshops, open field days and farmer learning exchange visits
 17. Climate proofing of water and sanitation infrastructure
 18. Training communities on hydroponics, conservation agriculture, agroforestry and agroecology
 19. Capacity building on impact of climate change on pastoralists communities
 20. Debunking climate change conspiracy theories
 21. Use of biogas and greenhouse food production
 22. Weather and climate data analysis for climate change detection
 23. Training in regenerative forestry-Carbon sequestration
 24. Agroforestry and climate smart agriculture extension
 25. Providing training of trainers on climate smart livestock practices and capacity building visits for technical staff interested in MRV of livestock systems
 26. Online activities and collaborating with climate change and wildlife conservation activities
 27. Dialogue on traditional knowledge on adaptation and mitigation
 28. Community extension and awareness activities
 29. KCSAP and NARIGP two counties value chain development



Figure 14: Some proposed learning activities by CSOs

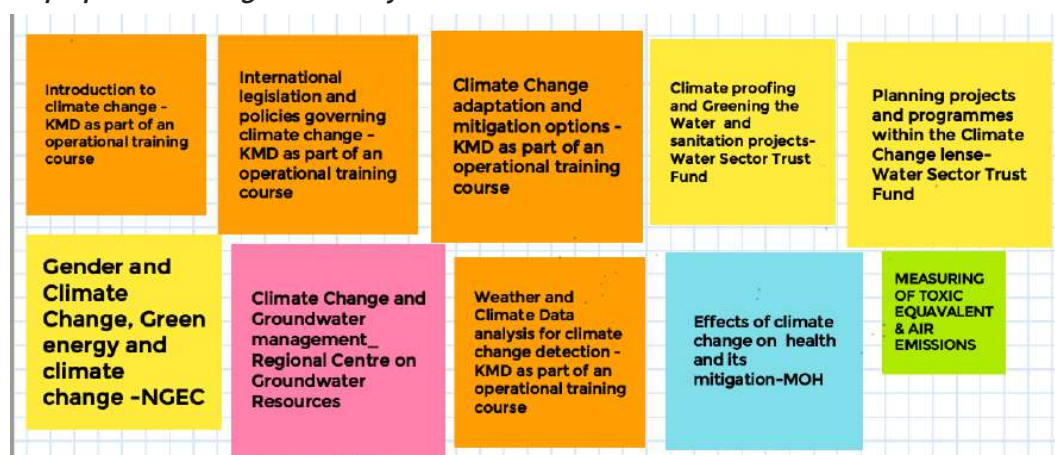


Figure 15: Some of the learning activities provided by government agencies

4.8 Climate Change Topics Offered

Basics of climate change science is the most offered climate change topic by training institutions at 79% followed by climate change and disaster risk management at 68%.

Adaptive and climate-resilient decision making and renewable energy training are offered by 65% of the training institutions while role of indigenous knowledge in climate change adaptation and gender and climate change are offered by 62% and 59% of the training institutions respectively.

Green economy, growth and jobs is the least offered climate change learning activity at 22% followed by development of a climate investment plan at 23%.

Carbon markets and Clean Development Mechanism and National and County Climate Change Funding are offered by 24% of the institutions while Greenhouse Gas inventory development is offered by 26% of the institutions.

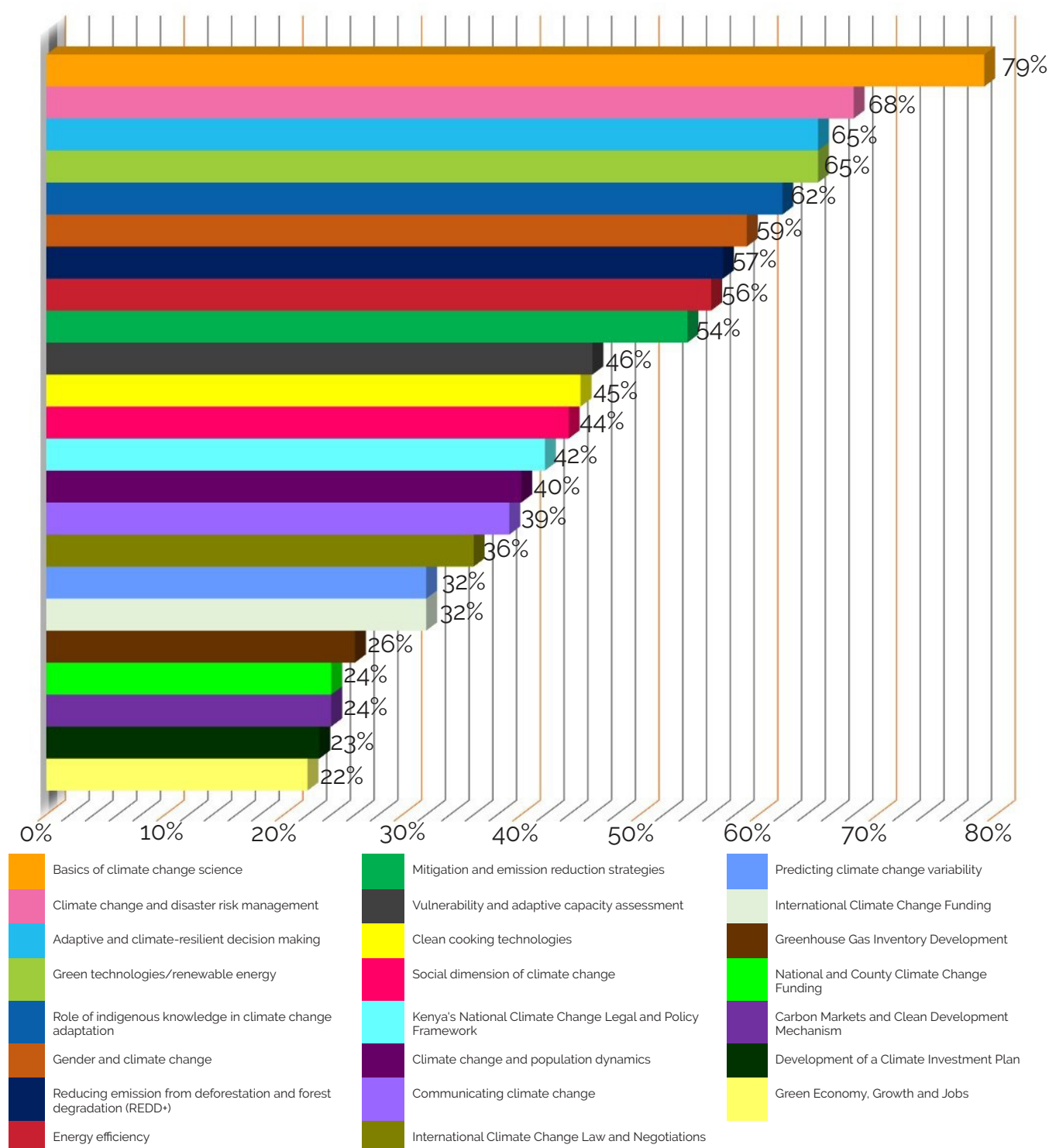


Figure 16: Climate change learning topics offered by institutions

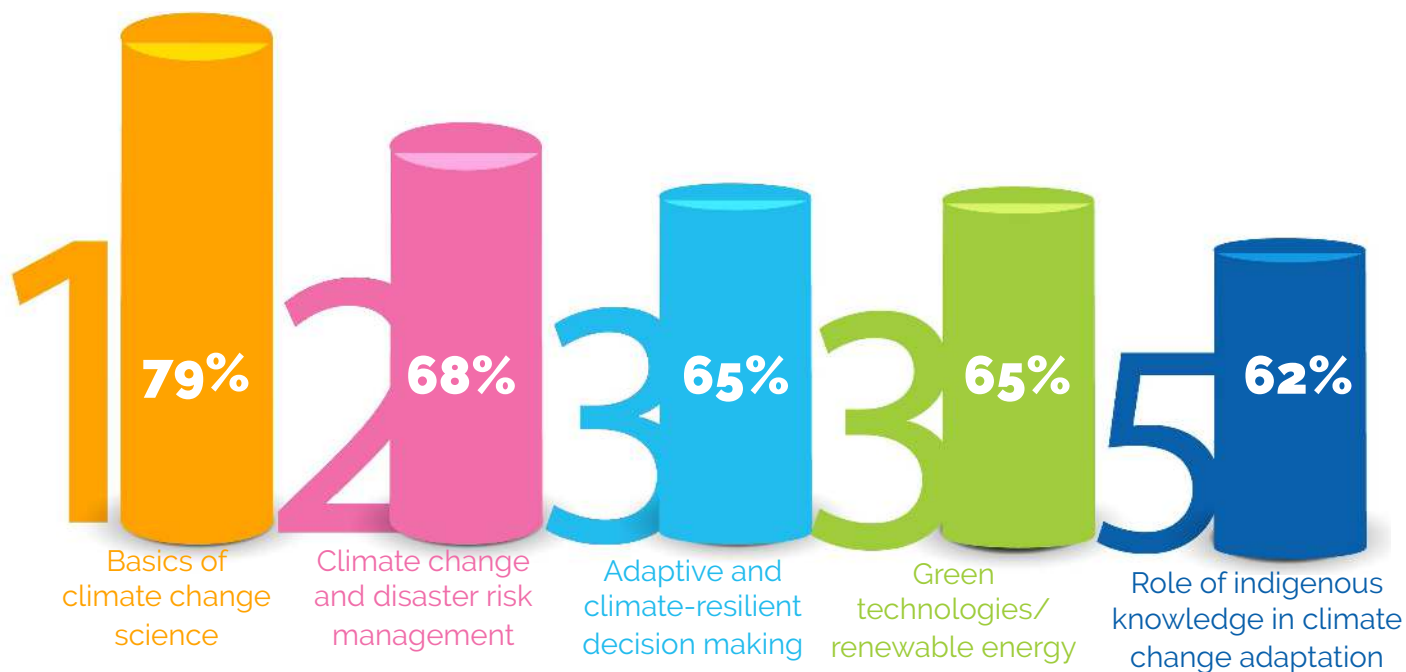


Figure 17: Top 5 most offered climate change learning topics by institutions

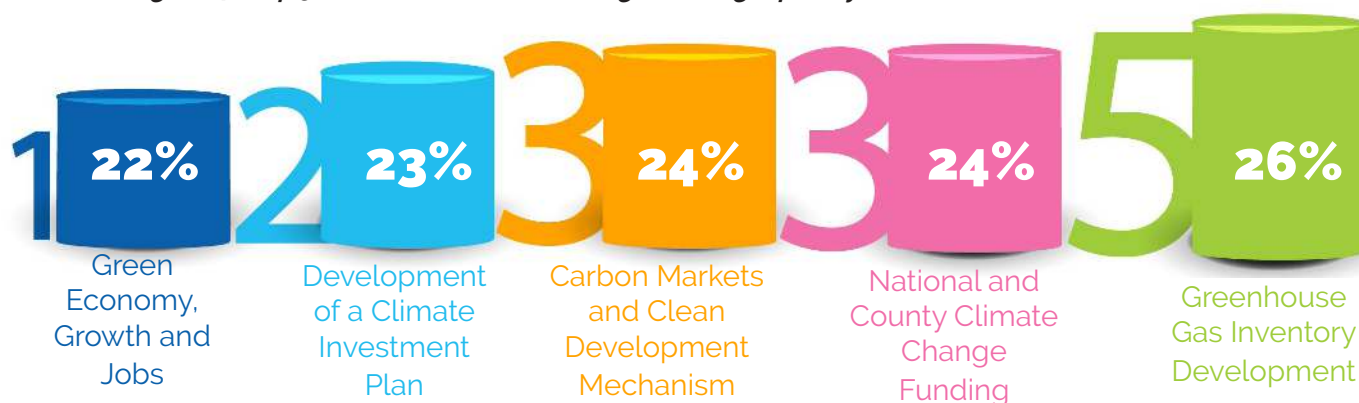


Figure 18: Five of the least offered climate change learning topics by institutions



4.9 Planned Climate Change Learning Topics

When it comes to climate change related learning activities that institutions plan to offer in future, climate change and disaster risk management lead at 72% followed by clean cooking technologies at 67%.

Future interest on social dimension of climate change and gender and climate change as a learning activity is at 65% each while role of indigenous knowledge in climate change adaptation and green technologies/renewable energy each have 64%.

Climate change and population dynamics has the lowest future interest as a climate change topic at 42% followed by development of a climate investment plan and National and County Climate Change Funding at 45%.

Future interest in offering learning on carbon markets and Clean Development Mechanism is at 47% while interest in reducing emission from deforestation and forest degradation (REDD+) is at 48%.

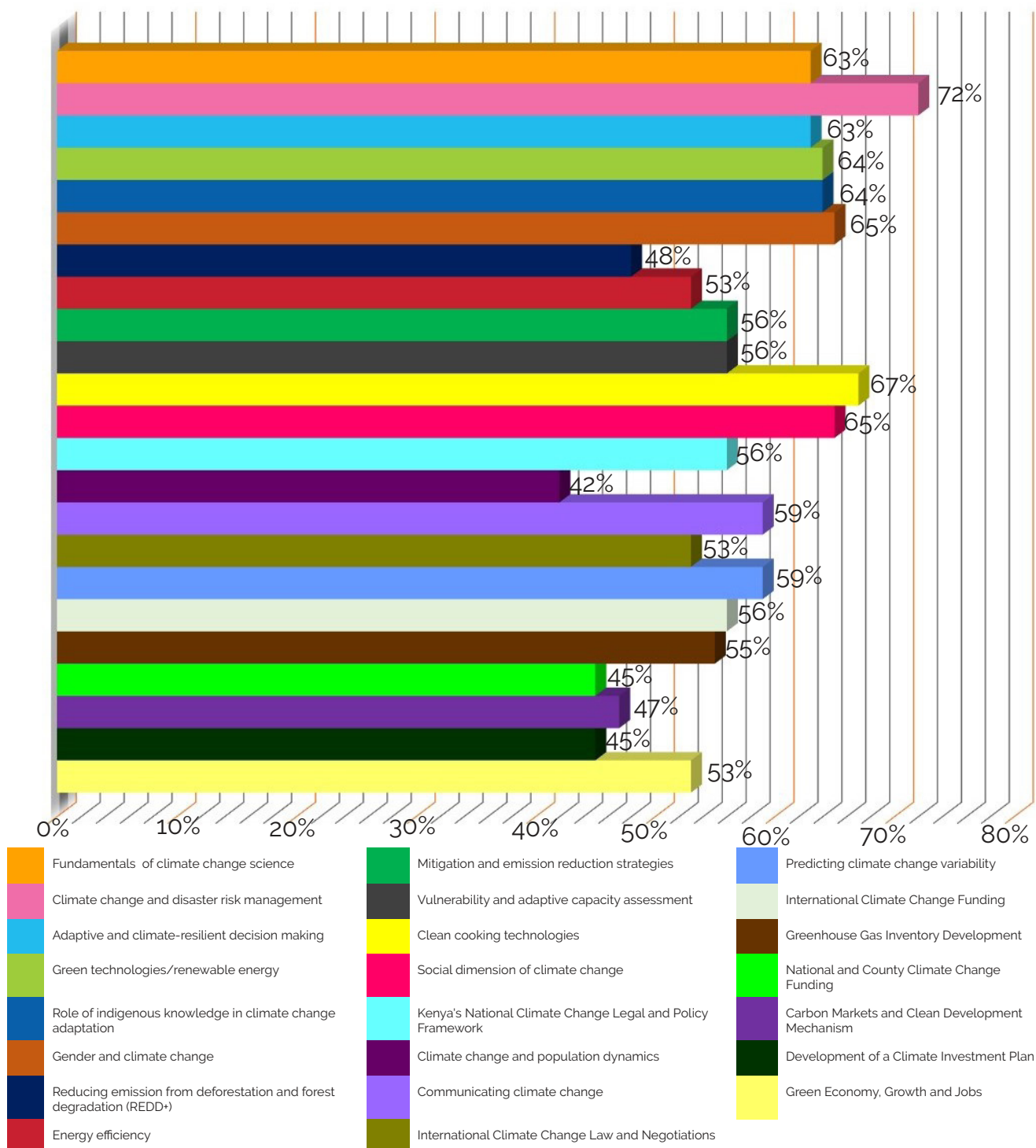
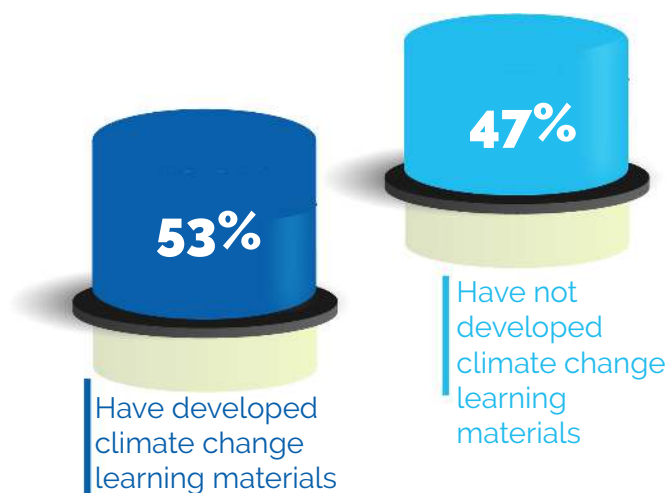


Figure 19: Planned climate change learning topics to be offered in future by institutions

4.10 Climate Change Learning Materials

When asked whether they have developed learning materials, training modules, e-learning courses relevant to climate change, 53% of the respondents said that they have while 47% have not developed any climate change learning material.



Some of the materials include

- A training manual on agroecology for climate change adaptation
- Training Manual on Protocol for a Tier 2 approach to generate region-specific enteric methane emission factors (EF) for cattle kept in smallholder systems <https://bit.ly/3biUIMe>
- Training Manual on Protocol for generating region-specific Tier 2 emission factors for methane (CH₄) and nitrous oxide (N₂O) emissions from cattle manure <https://bit.ly/2XpabIM>
- The Kenya School of Agriculture is currently developing curriculum and materials under sponsorship from GIZ
- Ecological Land use management Practices handbook
- Medicinal trees, Muunda Museo, Utooni Development Organization case study on climate change mitigation
- Adopted from the trainings from the partner institutions like KARLO and field experiences the latter is built of long time in the field.
- Climate change adaptation training manuals

- REDD+ training manuals for communities
- Curricula on: Community-Based Climate Risk Management, Environmental and Social Safeguards, Social Risk Management, Climate Change Policy Formulation, Green Growth, and Circular Economy
- County Climate Change Fund guidelines
- Curriculum Designs for Early Years Education; Environmental Activities
2.Primary Education; Social Studies
3.Senior Secondary: Geography
5.Teacher Education: Environmental Education and Social Studies
- Climate change developed by partnership with UNDP and FAO e-learning portal at KSG
- Climate Smart Agriculture Training manual

4.11 Climate Change Learning Expertise

The survey on expertise that trainers working in different institutions have concerning climate change topics indicates that institutions in Kenya have trainers with diverse expertise in the five prioritised sectors and three cross cutting themes.

Some of the available expertise are in the following climate change topics

1. Climate finance
2. Agroecology
3. Weather and climate data analysis
4. Science of climate change
5. Methods in climatology
6. Climate change adaptation
7. Gender and Climate change
8. Natural resources and disaster management
9. Renewable energy
10. Climate Resilient Agriculture
11. Integrated pest management
12. Climate smart agriculture
13. Environmental science

14. Climate Change Adaptation
15. vulnerability assessment
16. Environmental impact assessment
17. Permaculture
18. Social dimensions of climate Change
19. Ecological organic agriculture
20. Climate risk management and food security
21. Indigeneous Technical Knowledge (ITK)
22. Relation between environment, water and ecosystems
23. Climate change legal, policy framework and communications
24. Curriculum content on climate change
25. Gender and climate change
26. Climate governance
27. Agriculture management
28. Agroecosystems and environment
29. Wildlife management
30. Agriculture and human ecology extension
31. Climate change adaptation and mitigation technologies
32. Human Dimensions and Health
33. Ecological justice
34. Impacts of climate change
35. Environmental management

4.12 Climate Change Learning Activities/Courses Coverage

The curricular developed by the Institute aims at equipping learners with knowledge, skills, attitudes, and values on climate change which can be applied in real-life situations- Winrose Rono, Kenya Institute of Curriculum Development (KICD)

Some of the learning activities/courses where climate change related topics are covered include;

Courses

1. Online Bsc and Msc in environmental resource management
2. MSc in Climate Change Science-UoN
3. Master of Climate Change Adaptation (MCCA)- UoN
4. PhD in Climate Change- UoN
5. MSc in Energy Technology- JKUAT
6. MSc in Environmental Legislation-JKUAT
7. MSc in Climate Change and Governance- About to start at JKUAT
8. Certificate Courses in ESIA,solar panel installation and biogas-JKUAT
9. Masters in Environmental Studies and Geography- Garissa University
10. Natural Resource Management and Food Security- Garissa University
11. MSc in Climate Change and Sustainability- Kenyatta University
12. MSc in Renewable Energy Technology- Kenyatta University
13. Bachelor of Science in Climate Change and Development With IT - Maseno University
14. Large Utility Scale Solar PV installation Course- Riara University
15. MSc. In Geothermal Energy Technology- DeKUT
16. Postgraduate Diploma in Geothermal Energy Technology - DeKUT
17. Bachelor of Science in Renewable Energy Technology and Management - Jaramogi Oginga Odinga University for Science and Technology
18. Bachelor of Science in Water Resources and Environment Management - Jaramogi Oginga Odinga University for Science and Technology
19. PhD in Dry Land Agriculture and Resource Management - Egerton University
20. MSc in Dry Land Agro Pastoral Systems - Egerton University

21. Bachelor of Technology in Renewable Energy - Kirinyaga University
22. Bachelor of Water Irrigation and Environmental Engineering - Kirinyaga University
23. Bachelor of Technology in Water sanitation and Habitat Irrigation - Kirinyaga University
24. Bachelor of Technology in Renewable Energy and Environmental Physics - Technical University of Mombasa
25. PhD in Dryland Resource Management - South Eastern Kenya University
26. Ph.D in Environmental Management - South Eastern Kenya University
27. Master of Science in Rangeland Resources Management - South Eastern Kenya University
28. Master of Science in Integrated Water Resources Management - South Eastern Kenya University
29. Bachelor of Science in Meteorology - South Eastern Kenya University
30. Bachelor of Science in Agroecosystems and Environmental Management - South Eastern Kenya University
31. Bachelor of Science in Hydrology and Water Resources Management - South Eastern Kenya University
32. Ph.D in Natural Resources - Kisii University
33. Master of Science in Natural Resources - Kisii University
34. MSc in Climate change and Adaptation - Masinde Muliro University of Science and Technology
35. Ph.D in Natural Resources Management - Masinde Muliro University of Science and Technology
36. Bachelor of Science in Renewable Energy Technology - Masinde Muliro University of Science and Technology
37. Bachelor of Science in Forestry Ecosystem Management- Maasai Mara University
38. Master of Agroforestry - University of Kabianga
39. Bachelor of Science in Agroforestry and Rural Development - University of Kabianga
40. Bachelor of Science in Energy and Environmental Technology - Mount Kenya University
41. Bachelor of Science in Environmental Science and Resource Management - Meru University of Science and Technology
42. Bachelor of Science in Environment and Natural Resource Management - Africa Nazarene University
43. Short Courses in Solar PV (T1,T2), T3 Grid Tie and T3 Hybrid - Strathmore University Energy Research Centre
44. Short Courses in Energy Management and Solar Water Pumping - Strathmore University Energy Research Centre
45. Diploma in Rangeland Management - South Eastern Kenya University
46. Certificate and Diploma in Climate Change Adaptation- Maseno University
47. Diploma in Agroforestry - University of Kabianga
48. Certificate in Renewable Energy Technologies - University of Eldoret
49. Certificate in Application of Solar Energy and Photovoltaics - Kenya Coast National Polytechnic
50. Corporate and Professional Short courses (Energy, Solar PV (T1, T2, T3)- Institute of Energy Studies & Research (IESR)
51. Certificate in Solar PV T1, T2- Sigalagala National Polytechnic
52. Certificate in Solar Installation Technology- Kiirua Technical Training Institute
53. Certificate in Solar PV Installation - Wote Technical Training Institute
54. Certificate in Solar PV Training- Runyenjes Technical and Vocational College
55. Certificate in Water Technology- Masai Technical Training Institute
56. Certificate in Solar Installation - Michuki Technical Training Institute
57. Certificate in Energy Efficiency in Electrical Utilities for Commercial and Industrial Systems - The Institute of Energy Studies and Research
58. Certificate in Energy Management Essentials - The Institute of Energy Studies

and Research

59. Certificate in Small Hydro Power Plant Fundamentals, Operations and Maintenance - The Institute of Energy Studies and Research
60. Certificate in Wind Energy Fundamentals (Design and Development Technology) - The Institute of Energy Studies and Research
61. Certificate in Installation and Maintenance of Solar Power PV Systems T1, T2- The Institute of Energy Studies and Research
62. Certificate in Installation of Hybrid and Grid ne Power Systems T3 - The Institute of Energy Studies and Research
63. Diploma in Water Resources Technology and Management - Kenya Water Institute (KAWI)
64. Certificate in Water Resources Technology Management - Kenya Water Institute (KAWI)
65. Certificate in Solar Photovoltaic Grade 1,2,3 - Kenya Christian Industrial Training Institute (KCITI)
66. Certificate in Solar (PV) Installation for Technician - Nkabune Technical Training Institute
67. Certificate in Environment Impact Assessment and Environmental Auditing - Technical University of Mombasa
68. Certificate in Solar PV Installation Level 3 - Rift Valley Institute of Science and Technology
69. Certificate in Biogas Technology - The Kenya YMCA College of Agriculture and Technology (KYCAT)
70. Certificate in Solar Photovoltaic - PC Kinyanjui Technical Training Institute
71. Certificate in Disaster Mitigation and Preparedness - Kenya Institute of Development Studies
72. Certificate in Solar PVC installation - Siaya Institute of Technology
73. Certificate in Solar Power Installation Technician - The Nyeri National Polytechnic
74. Environmental impact assessment course
75. Operational Training Course for

Meteorologists

76. Climate Smart Agriculture Course offered to Students taking agricultural courses
77. SRM course where there is a topic on mainstreaming climate change in SRM
78. Permaculture design Certification Course

4.13 Integration of Climate Change Learning in Institutions

Some of the opportunities shared by respondents on how specific learning activities/courses in which climate change consideration could be integrated in their institutions include

1. Including climate change as one of the crosscutting issues in all courses offered
2. Using community capacity building as long ther (structured) learning activities
3. Development of curriculum and modules for climate change
4. Mainstreaming climate change in county planning
5. Having climate change as an entire diploma course
6. Ensuring that climate change is an independent department in institutions rather than a section of a department
7. Incorporate climate change learning in all public institutions
8. Development of community outreach programme to foster collaboration with communities



National Climate Change Resource Centre
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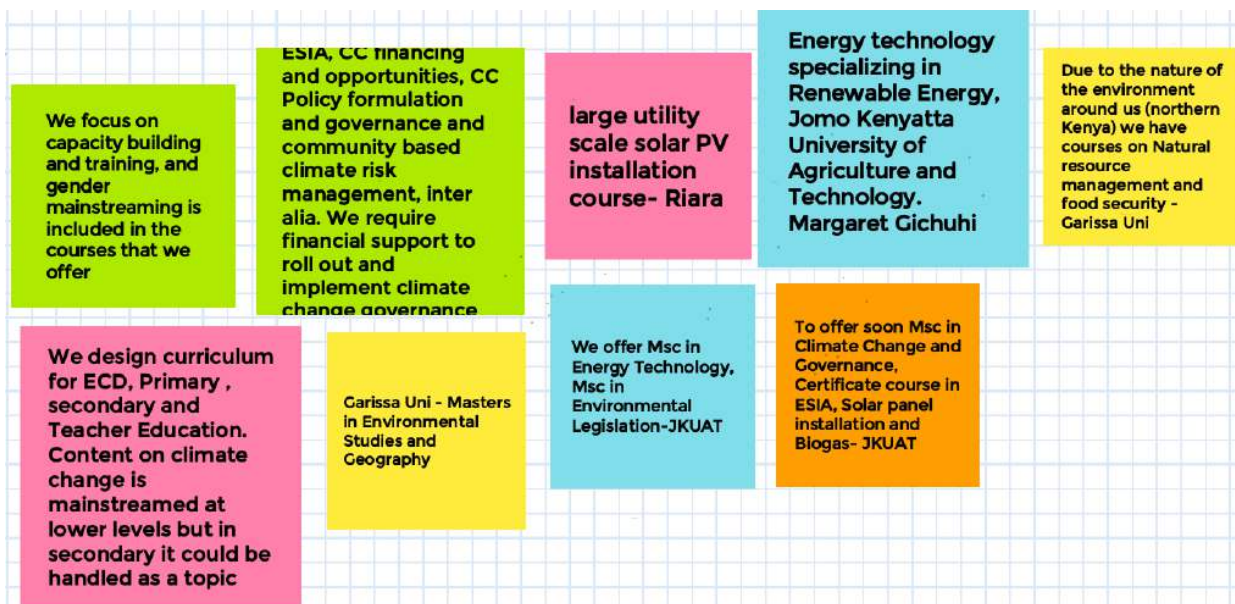


Figure 20: Some climate change learning topics/courses offered by institutions

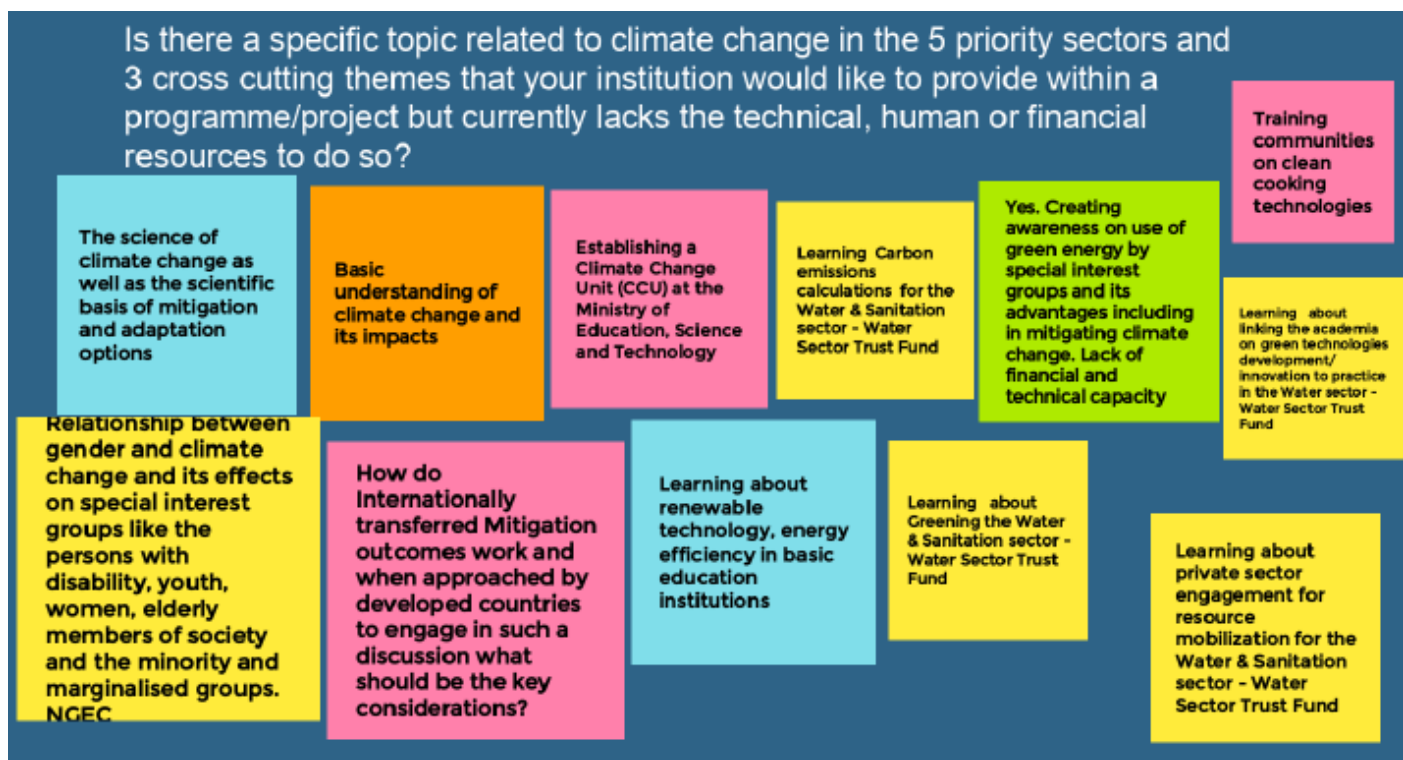


Figure 21: Some climate change learning topics that institutions plan to offer in future

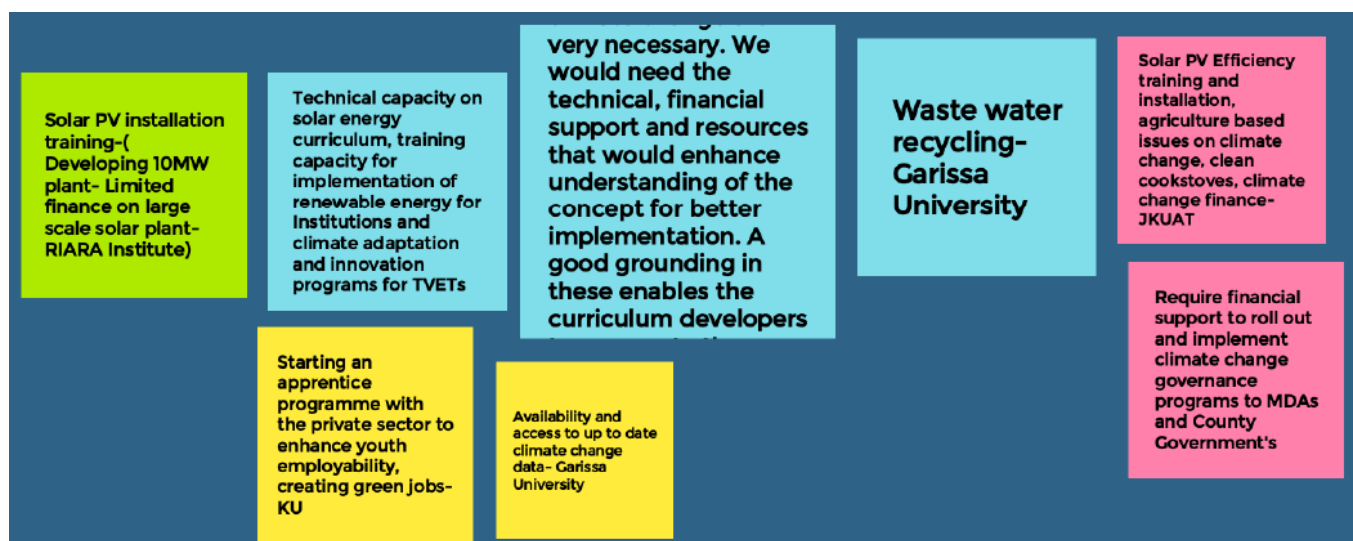


Figure 22: Some of the support required by institutions

4.14 Additional Issues Raised During FDGs

- 1 Address disinterest in students in taking up natural resource-based courses including environment, food security and climate change related courses at tertiary level
- 2 Introduce more experiential and hands-on climate change learning methods – *learning by doing*
- 3 Consider introducing or strengthening non formal programs such as Environmental clubs and societies, art, music, drama and science journalism including Eco Schools Programs
- 4 Strengthen linkages between academia and industry to address appropriate response to climate change
- 5 Enhance linkages between institutions to allow for sharing of relevant and current climate related data
- 6 Should climate change be a separate subject or infused within other subjects?
- 7 Introduce climate change themes during environmental events such as World Environment Days and World Biodiversity Day
- 8 Ensure that the Strategy doesn't limit actions to only the five priority areas and should include other key sectors such as the built environment and waste management among others
- 9 Funding mechanisms for implementing the strategy, sustainability and scalability framework should be built into the Strategy
- 10 Building capacity of the media to document and disseminate climate change stories
- 11 Engage with faith-based organizations to develop strategy to integrate climate change into faith based preaching and teaching to disseminate climate change messages to their congregations especially at grass root level
- 12 Develop coordination mechanism to engage Private sector to facilitate climate change learning

Chapter

05

Proposed Learning Actions

Education

1. Integrate climate change in the education system, emphasizing integration in existing curriculum for lower secondary grades 7, 8 and 9
2. Link research and academic institutions to climate change policy processes to provide scientific basis and promotion of climate change learning
3. Support the validation and finalization of the mainstreaming guidelines for integration of climate change in education curricula being developed by the Ministries of Environment and Education
4. Enhance the capacities of teachers and facilitators to teach climate change
5. Develop appropriate supporting supplementary teaching and learning materials
6. Leverage on non-formal and informal learning like creative arts, music and drama festival, school clubs, experiential learning among others
4. Engaging learners and community members on role of forest conservation in climate change and tree planting/nurturing campaigns
5. Sensitize and engage communities in the participation of blue economy related activities
6. Build sustainable partnerships, collaboration and synergies with existing and new initiatives

Agriculture

1. Promote adoption of climate smart agriculture
2. Strengthen communications systems on climate smart agriculture extension and agro weather issues
3. Increase awareness on climate change impacts on the agriculture sector
4. Promote the uptake of climate change related information for agriculture
5. Conduct capacity building in indigenous knowledge, livestock insurance schemes, early warning systems, early action, livestock management and breeding
6. Promote the up-scaling of climate resilient strategies/ technologies in fisheries and climate resilient fish varieties

Environment

1. Institutional strengthening of the CCD, the Climate Change Units and related institutions across sectors and counties as well as non state actor institutions
2. Strengthen early warning and tailor made climate information services through institutional strengthening of the Kenya Meteorological Department and other information users institutions
3. Capacity building of county level institutions for the efficient devolving of community forests

Energy

1. Develop guidelines on how to climate proof energy infrastructure using vulnerability risk assessments
2. Build capacity of various stakeholders on renewable energy and clean cooking technologies
3. Promotion of green cities and resource efficiency
4. Create awareness on the fuel economy

and electric mobility options, including exploring infrastructure needs for electric mobility

5. Build capacity of County Governments to develop energy plans
6. Capacity building on sustainable waste management in urban and rural areas especially slum areas and dumping sites

Water, Irrigation and Sanitation

1. Enhance capacity of institutions and bodies responsible for water, irrigation and sanitation on climate change impacts in the water sector
2. Conduct climate and risk assessments on water, irrigation and sanitation infrastructure
3. Mainstream climate change into water catchment management plans
4. Promote water efficiency (monitor, reduce, re-use, recycle and modelling)
5. Strengthen water resource monitoring and assessment for early warning and planning
6. Promote awareness on climate change impacts and the water sector

Capacity Building

1. Build the capacity of County Governments by strengthening of CCCUs and setting up functional CCCFs
2. Develop county adaptation plans for the counties with CCCFs
3. Build the capacity of land planners in using climate change future scenarios in land use planning
4. Build stakeholder capacities including vulnerable groups and PWDs to participate in, attract funding for, and report on climate change actions
5. Develop targeted specialised training programs to strengthen the capacity of various stakeholders to respond to climate change
6. Conduct capacity building on green jobs enterprises

Gender and Youth Engagement

1. Re-introduce the Kenya Climate Change Art and Essay Competition to reach and engage young learners
2. Promote gender responsive technologies and innovations in the private sector through capacity building
3. Create awareness for climate finance opportunities that women and youth can access
4. Build capacity for effective gender integration in NCCAP and NDC implementation
5. Encourage youth innovation through climate change outreach activities
6. Develop a national gender and inter-generational responsive awareness plan

Public Awareness

1. Operationalise the National Climate Change Resource Centre (NCCRC) with a robust Kenya Climate Change Knowledge Portal
2. Promote the establishment and operationalization of Community Education, Business and Information Centres (CEBICs) in select counties
3. Develop clear, simple localized and contextualized climate change IEC materials including posters, animations and documentaries
4. Undertake sensitization and awareness raising climate change public awareness campaigns across sectors and governance levels
5. Develop and operationalise a public awareness and engagement strategy that highlights outreach to politicians and media
6. Integrate indigenous traditional knowledge into climate change learning especially for communities most affected by the climate crisis



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